

Day 13 Observation Guide: Area of a Triangle

OBSERVER GUIDE

Grade 6 Mathematics · Professor Xavier Honablué, M.Ed. · For visiting guests and observers

Purpose of This Guide

Thank you for observing our Grade 6 mathematics classroom today. Days 8 through 12 built one durable idea: a leaning parallelogram is a rectangle in disguise, so its area is base $\times$ perpendicular height. Day 13 spends that capital. Students cut a parallelogram along a diagonal, discover they are holding two identical triangles, and derive $A = \frac{1}{2} \times \text{base} \times \text{height}$ for themselves rather than receiving it as a rule to memorize.

This guide explains how today's lesson was designed, maps it to the i-Ready Classroom Mathematics lesson it is drawn from, names the student errors we expect and how we plan to meet them, and links every material and research source used to build it.

The Design Decision: Derive, Don't Announce

The formula $A = \frac{1}{2}bh$ can be delivered in thirty seconds. We spend the first fifteen minutes not delivering it. Each student gets a paper triangle and makes a second identical copy; rotating the copy 180° and sliding it against the first always produces a parallelogram with the same base and the same height. The $\frac{1}{2}$ is then not a rule — it is the visible fact that our triangle is half of a shape whose area the class already proved on Day 8.

This ordering matters for the error we know is coming. Students who memorize " $\frac{1}{2}$ base times height" drop the $\frac{1}{2}$ under pressure, because nothing in their memory explains it. Students who watched two triangles become one parallelogram have something to reason back to when they are unsure — which is exactly the multi-strategy habit the IES problem-solving practice guide recommends.

The Height Problem Returns — On Purpose

Day 8 and Day 9 taught students to reject the slanted side. Triangles raise the stakes: a triangle has three possible bases, each with its own height, and on an obtuse triangle the height falls outside the figure and must be drawn to an extended base. Today's practice set deliberately includes an obtuse triangle labeled with base, height, and slanted side at once, plus a right triangle where the two legs do the work. Observers should expect to hear students arguing about which number is the height. That argument is the lesson.

i-Ready Classroom Mathematics Alignment

Today's lesson is drawn from i-Ready Classroom Mathematics, Grade 6, Unit 1, Lesson 2 ("Find the Area of a Triangle"), continuing the unit sequence begun with Lesson 1 on Days 8–9 and reviewed on Days 11–12.

i-Ready content (Grade 6, Unit 1)	Where it appears today	Standard(s)
Lesson 2, Session 1 · Explore — two copies of a triangle form a parallelogram	Opening hands-on derivation; Learn section 1 of the Lesson & Practice packet	6.G.A.1, MP.7
Lesson 2, Session 2 · Develop — $A = \frac{1}{2}bh$ with right, acute, and obtuse triangles	Worked Examples 1–2 (including the obtuse triangle with an outside height)	6.G.A.1, MP.6
Lesson 2, Session 2 · Try It / Apply It — missing base or height	Worked Example 3 and practice items 5–6	6.G.A.1, 6.EE.B.7
Lesson 2, Session 3 · Refine — composite figures and error analysis	Worked Example 4 (rectangle + triangle) and practice items 8–9	6.G.A.1, MP.3

i-Ready content (Grade 6, Unit 1)	Where it appears today	Standard(s)
Lesson 2 · Math Journal and vocabulary (base, height, perpendicular)	Closing journal prompt; vocabulary is boxed in the packet	MP.6

Teacher note: session numbers follow the i-Ready Classroom Mathematics Grade 6 Teacher Edition. Confirm exact page numbers against your edition before distributing this guide to observers.

Lesson Flow — 55 Minutes

Min	Segment	What the class is doing	Look for
0-5	Do Now	Two parallelogram area problems from Day 12, one with a distractor slanted side.	Are students marking the perpendicular height before computing?
5-20	Explore (hands-on)	Cut and match two identical triangles into a parallelogram; state the $\frac{1}{2}$ relationship in students' own words.	Student language: "half of the parallelogram," not "the rule says."
20-35	Develop	Worked Examples 1-3: acute, obtuse with an outside height, and a missing-height problem.	Do students double the area before dividing when running the formula backwards?
35-48	Practice	Lesson & Practice packet, items 1-12, in pairs; teacher confers with 4-5 students.	Partner talk that names base and height explicitly.
48-55	Refine & close	Error analysis (item 8: the student who forgot the $\frac{1}{2}$) and the Math Journal prompt.	Can students explain why the $\frac{1}{2}$ belongs there?

Predictable Student Errors — and What We Do About Them

Error	Why it happens	Our move today
Forgetting the $\frac{1}{2}$	The formula was memorized as a string of symbols with no picture behind it.	Send the student back to the two-triangle model; ask which shape their answer describes.
Using the slanted side as the height	It is the longest, most visible number in the drawing.	Same move as Day 9: trace the right-angle mark with a finger; compare slant vs. height lengths.
Obtuse triangle: "there is no height"	The perpendicular segment falls outside the triangle.	Extend the base with a dotted line on the board and re-mark the right angle.
Missing base/height: dividing before doubling	Students undo the multiplication first and forget the $\frac{1}{2}$.	Write $2A = b \times h$ as its own line before dividing.

What an Observer Can Look For

- Students using the words **base**, **height**, and **perpendicular** without prompting.
- At least two different strategies visible in the room for the same problem (formula, doubled-copy picture, decomposition on grid paper).
- Every student's notebook showing hand-drawn figures with a dotted height and a right-angle mark.
- Error analysis that diagnoses a mistake rather than only marking it wrong.
- Units labeled as **square** units on every answer.

Today's Materials — Direct Links

Every item below is also one click away on the launcher: cpamathlauncher.streamlit.app.

Material	Link
Lesson & Practice packet (12 problems + full solutions)	Day13_Lesson_Packet.pdf
Anchor chart — Area of a Parallelogram (prior day, still posted)	Anchor_Chart_Area_of_a_Parallelogram.pdf
Day 12 review sheet (warm-up source)	Day12_Review_Sheet.pdf
Homework — IXL, Grade 6 skill "Area of triangles" (90% target)	ixl.com/math/grade-6/area-of-triangles

Standards Addressed

Code	Standard	Where it shows up today
6.G.A.1	Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes.	The entire lesson; composite figure in Example 4.
6.EE.B.7	Solve real-world and mathematical problems by writing and solving equations of the form $px = q$.	Missing base and missing height problems (items 5–6).
MP.3	Construct viable arguments and critique the reasoning of others.	Error-analysis item 8 and partner justification.
MP.6	Attend to precision.	Choosing the perpendicular height; labeling square units.
MP.7	Look for and make use of structure.	Seeing a triangle as half of a parallelogram.

Research & Further Reading

Every hyperlink below opens the source directly. These are the sources used to build today's lesson.

Source	Why it is here
Common Core State Standards — Grade 6, Geometry (6.G.A.1–4)	The exact wording of the standards this unit teaches, including the "compose and decompose" language our lessons are built on.
IES / What Works Clearinghouse — Improving Mathematical Problem Solving in Grades 4–8 (full PDF)	The federal practice guide behind our daily routine: worked examples with steps, student error analysis, and multiple solution paths.
Curriculum Associates — i-Ready Classroom Mathematics	The publisher page for the curriculum this unit follows (Try–Discuss–Connect instructional routine).
i-Ready Family Center	Free family-facing explanations of each lesson — useful to hand to parents who ask how to help at home.
YouCubed at Stanford — open math tasks	Stanford Graduate School of Education tasks used for our visual and multi-strategy work.
Illustrative Mathematics — Grade 6, Unit 1, Lesson 9: Formula for the Area of a Triangle	Open-licensed lesson using the same two-copies derivation; good source for extra classroom tasks.
GeoGebra — Area of Parallelograms and Triangles	Projectable applet: drag a vertex and watch base $\times$ height stay fixed while the shape leans.
Khan Academy — Triangle area proofs	Short video for students who were absent or want the derivation again at home.

Questions during or after the observation are welcome. Every material listed above is public on the launcher, and the Lesson & Practice packet includes the full worked solutions, so families can follow the same steps at home.